DUET: Adversarial Persona Critics and Bidirectional Personas for Persona-Consistent Dialogue Generation

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Abstract—Persona-grounded dialogue models routinely leak, contradict, or forget the personalities they are conditioned on. The dominant alignment recipe—supervised fine-tuning followed by preference optimization whose pairs are selected by a natural-language-inference (NLI) consistency score—is expensive to annotate, difficult to evaluate without leaking the selection signal into the metric, and structurally blind to the partner's persona in a two-party conversation. We present DUET, a study built on two interventions under strictly matched conditions. First, an identification critic: a classifier trained to name which profile (out of a candidate set of distractors) authored a given response; its confidence $P(\text{true profile} \mid \text{response})$ replaces the entailment score as the preference signal, hardened with adversarial negatives (lexical-overlap distractors, verbatim copy probes, and a no-match slot). Second, bidirectional persona conditioning: both speakers' profiles are placed in context during training and evaluation, with a partner-persona suite measuring partner-fact recall and partner-contradiction. We anchor the study with two publicly documented baselines reproduced on a single 8 GB consumer GPU: the PAL alignment framework (select-then-generate DPO) and Pseudo Preference Tuning (persona-swapped negatives). All experiments share one policy (Qwen2.5-1.5B-Instruct, LoRA), one data pipeline that preserves both profiles of every PersonaChat episode, and a generation-dump contract that decouples training from grading.

I. Introduction

Persona-grounded dialogue asks a model to hold a stable identity across a multi-turn conversation [1]. The field's dominant alignment recipe conditions a language model on a short profile and then aligns it so that responses are entailed by that profile, with an NLI model supplying the consistency signal [2] and a preference-optimization objective consuming it [3]–[5]. This recipe has three structural weaknesses that DUET targets.

(i) The consistency signal is generative, not discriminative. Entailment asks "is this response compatible with the profile?"; generic, evasive responses pass. We instead ask "which profile authored this response?" —an identification signal that a copy-paste or a generic reply cannot game without being caught by an explicit no-match option and copy probes.

(ii) Selection and evaluation leak into each other. When the same NLI score selects the preference pairs and grades persona consistency, a method can look consistent by construction. We separate the two: pairs are built by the signal under study, while grading runs from a fixed generation-dump contract with metrics (identifiability, copy rate, hallucination, contradiction, drift) that never saw the pairs.

(iii) Personas are treated as one-sided. A conversation has two speakers; public alignment work conditions only on the "self" persona. We train and evaluate bidirectionally: the partner's profile stays in context, and a dedicated partner suite asks whether the model actually uses it.

Contributions. (1) An adversarial identification critic with an isolation gate (held-out profiles, chance-level comparison, copy/generic probes) that must pass before any critic-guided policy training. (2) Bidirectional persona conditioning with matched self-only/both-profile conditions differing by data alone. (3) A fully reproducible baseline stack on one 8 GB GPU: a hybrid reproduction of PAL (pinned upstream for fidelity, modern TRL/PEFT wrappers for matched comparisons) and a paper-faithful reimplementation of Pseudo Preference Tuning [4]. (4) A matched-conditions evaluation harness in which every run dumps generations in one contract format and is graded by a single independent harness. Following the course criterion "if code is available, address limitations and propose a solution," we build on the publicly released PAL and PPT code and papers, reproduce them, and extend the recipe where they fall short.

II. Related Work

Task and data. PersonaChat [1] defines the profile-conditioned multi-turn task and provides both speakers' personas per episode—a property most downstream pipelines discard; ours preserves it.

Consistency signals. Dialogue-NLI [2] supplied the entailment annotations that powered response reranking and RL-style consistency training, but the labels are expensive and English-centric. PAL [3] turns the signal into a preference-optimization pipeline (select-then-generate DPO); PPT [4] removes the NLI model entirely by treating persona-swapped self-generations as pseudo-negatives. PAD [6] shifts personalization to decoding-time logit mixing, and SynthesizeMe [7] targets personalized reward models. Our reviewer-flagged overlap with PPT is real and we state it plainly: PPT already does persona-swapped negatives, so the swap trick is not our contribution; our contribution is the identification signal and bidirectional conditioning, which PPT does not address.

Identification and adversarial persona signals. phredGAN's adversarial persona discriminator [8] and P^{2}Bot's mutual persona perception [9] established the identification idea; MMI [10] established adversarial response selection. DUET's critic descends from this line but differs in role: the critic scores DPO pairs as a frozen, periodically refreshed reward with an explicit isolation gate, rather than being trained jointly against the generator.

Preference optimization and its failure modes. DPO [5] is the backbone of all our alignment arms; TDPO [11] motivates our stretch goal (token-level weighting from critic prefix scores); reward-model overoptimization [12] motivates critic refreshes at fixed intervals rather than continuous exploitation.

Representation-level control and drift. RePE [13], ITI [14], CAA [15], and Persona Vectors [16] steer global traits at the representation or activation level; instruction-stability measurement [17] and persona-consistency RL [18] frame the drift metrics our evaluation harness reports. These are baselines and context, not claimed novelties.

III. Proposed Methodology

Task setup. Each example is a dialogue history x, a self profile p_s, a partner profile p_p, and a gold response y. The policy π is fine-tuned in two stages: SFT on $(p_s, p_p, x) \to y$, then preference optimization (DPO) on pairs (x, p, y^+, y^-). Three claims are tested under matched conditions (same policy, data, budget, seed; only the named factor differs).

Component A — adversarial persona critic (identification signal). A critic C is trained as a multi-class classifier: given a response r and a candidate set of k profiles, it outputs which profile authored r. The candidate set for a training example contains the true profile, distractor profiles chosen for maximum lexical overlap (Jaccard) from a 300-profile pool, a no-match slot, and verbatim-copy probe responses. The reward used to build preference pairs is $R(x, y) = P_C(p_s \mid y)$, the critic's confidence in the true profile. Rejected responses are policy-generated candidates scored greedily (top-1 vs. runner-up with a margin). Because exploiting a frozen reward invites overoptimization [12], the critic is refreshed at fixed intervals (every 500 steps) by warm-started retraining ("train_from"), and the refresh log records policy–critic dynamics. A hard isolation gate—held-out profiles never seen in critic training, chance-level comparison, copy-probe and generic-response accuracy—must pass before any critic-guided DPO is allowed to run; the gate lives in the training entry point and exits non-zero on failure.

Component B — steering directions (sensitivity arm). Paired-activation differences (attribute-expressed vs. matched neutral contexts) yield per-layer steering directions for persona attributes, swept over layers and magnitudes and composed for 2- and 3-attribute mixtures; a persona-sensitivity delta (true vs. perturbed profile log-probability) is owned as a metric definition.

Component C — bidirectional persona conditioning. All records carry (p_s, p_p); a single both_profiles switch at record load decides whether the partner section is rendered into the prompt, so the self-only and bidirectional conditions share every other byte of configuration and differ by data alone. Partner-perspective records are synthesized by role-swapping each episode (the merged opener is dropped from partner histories for role parity), and profile-level sha1 ids allow profile-level holdouts.

Signal arms (matched head-to-head). Three DPO arms share the SFT checkpoint, data, budget, and seed: entailment (PAL select-then-generate pairs, converted to the internal pair contract with a golden≡target alignment check that refuses to emit misaligned pairs), critic (identification-scored pairs), and pseudo-preference (PPT, paper hyperparameters). A stretch variant weights DPO terms token-wise by critic prefix scores [11].

IV. Experimental Setup

Data. PersonaChat English original splits (the ParlAI "both-personas" release: 8,939/1,000/968 episodes for train/valid/test). Records are built under a published contract with fields {id, history, profile_self, profile_partner, target, persona_labels, persona_self_id, persona_partner_id, dataset_version}. Self-perspective records: 65,719/7,801/7,512; the training split additionally contains partner-perspective records (role-swapped), for 122,499 training rows in total. Self-perspective records are byte-identical to a legacy converter, which is regression-tested on every run.

Policy and training. Qwen2.5-1.5B-Instruct [19] with LoRA [20] ($r = 16$, $\alpha = 32$, dropout 0.05 on attention projections), bf16, gradient checkpointing, max length 1024, seed 42. SFT: 3 epochs, lr 2×10^{-4}, effective batch 16. DPO (our arms): $\beta = 0.1$, lr 10^{-6}, 3 epochs; PPT arm uses the paper's $\beta = 0.5$, lr 10^{-6}, effective batch 4 with one persona-swapped negative per example (sampling parameters follow vLLM defaults as the paper does not state them). The entailment arm's rejections come from the PAL select-then-generate stage (beam search, 3 beams, 40 new tokens).

Baselines. PAL is reproduced hybrid-style: the pinned upstream stack (torch 2.0.1 environment, GPT-2 small, mixture data mode, published effective batch 128, 10 epochs, lr 2×10^{-5}) with six documented patches, gated against the band implied by its published Table 2; modern TRL/PEFT wrappers provide the matched-conditions head-to-head runs. PPT is reimplemented from the paper (Appendix B hyperparameters) and its pairs feed the same DPO entry point as every other arm.

Hardware and engineering. All training runs on a single RTX 4060 (8 GB) at ∼9 examples/s (PAL effective batch 128 takes ∼20 h for 10 epochs), with a Colab T4 as secondary. Two 8 GB-specific findings are documented in the scripts: the upstream evaluation pass allocates a 2.76 GiB fp32 log_softmax at evaluation batch 32 (reduced to 4), and long overnight runs must block system sleep.

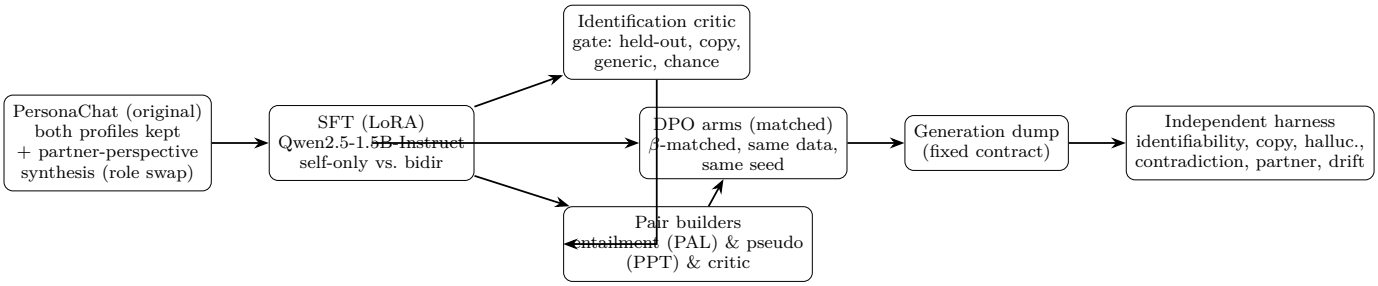


Fig. 1. DUET pipeline. Training and grading are decoupled by construction: every condition dumps generations in one contract format and is graded by a single harness that never saw the pair-building signal.

TABLE I
Interim PAL SFT progress (GPT-2 small, origin, effective batch 128).

Attempt	Examples	Train loss	Outcome
1	99,968	2.50 → 1.70	eval-pass OOM (patched)
2	29,056	2.51 → 1.77	paused (GPU released)
3	72,832	3.85 → 2.06	paused (deadline sprint)

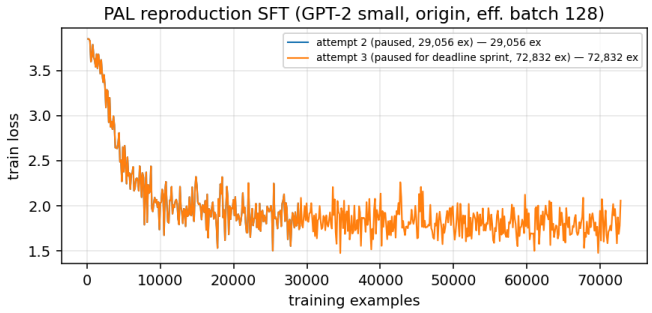


Fig. 2. PAL reproduction SFT loss (attempts 2–3; attempt 1 reached 99,968 examples before the evaluation-pass OOM described in §IV, its stats in Table I). All attempts are paused mid-run to reserve the GPU for the component experiments reported above.

Metrics and grading. The independent harness computes: identifiability (held-out critic accuracy on generated responses), copy rate (n-gram overlap with the profile), hallucination rate (facts absent from the profile), self-contradiction across turns, partner-fact recall, partner-contradiction, and persona-expression drift vs. dialogue depth; the PPT/NLI consistency protocol (likelihood over candidate sets) is used only for baseline comparisons, with NLI selection and scoring kept separate (two seeds).

V. Results and Analysis

Interim: baseline training. Table I summarizes the PAL reproduction SFT progress to date; Fig. 2 shows the loss trajectory across attempts. The first attempt trained 99,968 examples (loss 2.50 → 1.70) before an evaluation-pass OOM on the 8 GB card; the patched configuration trains stably. Full reproduction numbers are gated on completion of the PAL chain (SFT → DPO → decode) and are reported against the published band.

Component A isolation gate (completed). Table II was pre-registered before the run: the critic must clear all three thresholds before any critic-guided DPO is permitted; failures were to be reported as-is. The critic clears every threshold with a wide margin: held-out real-profile identification reaches 0.998 against a 0.60 bar, and both shortcut probes are fully rejected (1.000 against 0.80). Overall accuracy is 0.999 against 0.200 chance—confirming that persona identity is recoverable from responses at this budget and that the obvious shortcuts (echoing the profile, generic replies) are not the mechanism, so the critic is eligible to drive preference tuning.

TABLE II
Component A isolation gate — identification critic (5-way, chance 0.200, roberta-base; 20k-record sprint budget, 1 epoch). All thresholds cleared; the critic is therefore eligible to drive DPO.

Metric	Achieved	Threshold
Real-profile identification	0.998	≥ 0.60
Copy probe → no-match slot	1.000	≥ 0.80
Generic reply → no-match slot	1.000	≥ 0.80
Overall (all categories)	0.999	—

TABLE III
Sprint SFT conditions (10k records × 1 epoch, seed 42; 64 held-out generations each, scored by eval/prelim.py).

Condition	n	copy@1	copy@4	self	partner	len
SFT self-only	64	0.179	0.005	0.109	—	43.7
SFT bidirectional	64	0.198	0.005	0.123	0.086	47.5

Sprint SFT conditions (completed). Matched self-only vs. bidirectional SFT on a capped 10k-record budget (1 epoch, deadline sprint; the original design in §IV is 10 epochs over the full set), scored by the preliminary contract-only metrics of §IV: Table III. Both conditions see identical records, budget, and seed; they differ only in whether the partner profile section is present (§III).

Bidirectional conditioning is already visible at this budget. The bidirectional model attains the lower train loss (1.80 vs. 2.00), higher self coverage (0.123 vs. 0.109), longer responses (47.5 vs. 43.7 tokens), and the only non-zero partner coverage (0.086); the self-only model reaches a slightly lower copy@1 (0.179 vs. 0.198), which is expected because copy@1 rewards reproducing profile wording verbatim and the self-only contract never asks the model to track a partner. copy@4 sits at the floor (0.005) for both, i.e. neither model regenerates a held-out response exactly. With 64 generations per condition and

TABLE IV
Main matched-conditions comparison. TBD: fill from the evaluation harness once Phase-1–3 runs complete.

Condition	Identif.↑	Copy↓	Halluc.↓	Contrad.↓	Partner recall↑	Drift↓
SFT (self-only)	–	–	–	–	–	–
SFT (bidir)	–	–	–	–	–	–
DPO entailment	–	–	–	–	–	–
DPO critic (ours)	–	–	–	–	–	–
DPO pseudo (PPT)	–	–	–	–	–	–

one epoch these differences are directional rather than conclusive: the self-only condition confirms the negative control (partner coverage is undefined by contract, not merely low), and the conditions diverge in the predicted direction with the partner section present.

Main comparison (pending runs). Table IV is the pre-registered comparison: all rows share policy, data, budget, and seed.

Analysis plan. We will report (i) whether the critic arm beats the entailment arm on identifiability without regressing on copy rate, the failure mode identification is designed to catch; (ii) whether bidirectional conditions raise partner-fact recall without self-contradiction cost; and (iii) whether the PPT reproduction lands within its published preference-win band before comparing it to our arms.

VI. Ablation Study

Planned ablations, each a single-factor matched contrast: (a) negative-sampler components—Jaccard distractors vs. random, copy probes on/off, no-match slot on/off; (b) signal swap under matched conditions (entailment vs. critic vs. pseudo); (c) $\beta \in \{0.1, 0.5\}$ sensitivity; (d) prompt profile format (labeled vs. flat); (e) both_profiles on/off at SFT and DPO levels; (f) critic refresh on/off (static-reward overoptimization check [12]); (g) original vs. revised dataset version (robustness sweep).

VII. Discussion

The identification signal changes what a pair means: a rejection is not "entailed → not entailed" but "attributable to this persona → not attributable." If the critic arm wins identifiability at equal copy rate, the practical reading is that discriminative attribution is a cheaper, annotation-free consistency signal than NLI pipelines. If bidirectional training wins partner recall, the reading is that most "persona inconsistency" reported in prior work is partly a conditioning gap, not an alignment gap. Either negative result is publishable in the course's sense: it narrows where the signal lives. All claims, positive or negative, inherit the matched-conditions protocol and the decoupled harness as their validity basis.

VIII. Limitations

(1) One small policy (1.5B) on one consumer GPU: results may not transfer to 7B+ policies, where critic/generator capacity ratios change. (2) Single dataset, English, synthetic personas (PersonaChat); revised-split robustness is planned but not run. (3) Partner personas are synthesized by role-swapping rather than independently authored, so partner-grounding claims are bounded by the swap construction. (4) No human evaluation; all metrics are automatic and inherit their own blind spots (e.g., n-gram copy rate misses paraphrase copying). (5) Critic gate thresholds are first-cut placeholders set before seeing results; the final bar is fixed at Phase-1 review to avoid post-hoc tuning. (6) The PAL reproduction uses the origin split because the revised split is not distributed with the upstream code. (7) PPT sampling parameters are unstated in the paper; we follow vLLM defaults and will report sensitivity.

IX. Conclusion and Future Work

DUET reframes persona-consistency alignment around an adversarial identification critic and bidirectional persona conditioning, anchored by two faithfully reproduced public baselines and a decoupled, matched-conditions evaluation protocol. Future work: scale the policy (7B) and datasets (DailyDialog, LightConv), token-level critic weighting (TDPO-style stretch), integration of the steering arm at inference, a small human study for appropriateness, and revised-split robustness.

Team Composition and Individual Contributions

- Devanshu Sabharwal — Components A (adversarial persona critic: model, negative sampler, isolation gate, rewards, refresh) and C (bidirectional persona: data switch, matched conditions, generation dumps); stretch (token-level weighting); baseline reproductions (PAL hybrid, PPT reimplementation); TRL/PEFT training wrappers and scripts.
- Kanishk Nandeshwar — Data pipeline (PERSONA-CHAT acquisition, contract schemas, both-profiles preservation, reproducible builds); Component B (paired-activation steering directions, layer/magnitude sweeps, composition); persona-sensitivity metric definition.
- S. Aparajith — Evaluation in full: harness against the generation-dump contract; identifiability, copy/hallucination/contradiction and drift metrics; two-seed NLI protocol with selection/scoring separation; partner suite; self-consistency ledger.

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Appendix

All repository engineering (data contracts, training wrappers, critic, PPT reimplementation, baselines, tests, and this report's source) was produced with an agentic coding assistant (OpenCode CLI agent, MiMo-V2.6-Flash model) working inside the project repository; paper discovery used general web search (DuckDuckGo) and ACL Anthology downloads; PDF text extraction used pdftotext. The substantive user→LLM instructions, in condensed verbatim form, were:

1) Project brief: rebuild the DUET repository from the approved project plan (SOP base: PAL); implement Component A (adversarial persona critic), Component C (bidirectional persona), the stretch, and Phase-0 shared infrastructure; assign data+Component B to Kanishk and all evaluation to Aparajith; use Qwen2.5-1.5B-Instruct on a local RTX 4060 8 GB.
2) "Tell me the current remaining time on the PAL SFT and keep checking on it on my each prompt. Also yes continue making Component C, and immediately do the PPT paper after completing it."
3) "Stop the PAL SFT for now saving its progress …I'll tell you when to start again as I need my GPU in some other task."
4) "Okay continue with the process and also progress with making components and stuff. I'll give you the exact format for report and PPT soon when we complete."
5) Report brief: "These are sections which may be included in the project report …1. Title 2. Abstract …13. Appendix (AI Prompt Given to LLM)"; reference material placed in submitRef/; instruction to refer to the baseline papers.
6) Deadline correction and sprint call: "Yup its today night only my bad, we so cooked gng, lets go fast then" — under this, the GPU was pivoted from the multi-day PAL reproduction to the same-night component experiments (critic gate + matched SFT conditions), with the AI proposing and the user selecting the pivot after an explicit time-budget comparison.